i.e., it does not depend on the initial concentration.

- According to collision theory, chemical reactions occur as a result of collisions between the reacting molecules.
- Generally, the rate of a reaction increase with increasing temperature. However, there are very few exceptions. The magnitude of this increase in rate is different for different reactions. As a rough rule, for many reactions near room temperature, reaction rate tends to double when the temperature is increased by 10^0C .
- According to Arrhenius, activation energy of the reaction is the minimum energy that a molecule must have to posses to react.
- The rate of a reaction is affected by the following factors.
  1. Nature and state of the reactant
  2. Concentration of the reactant
  3. Surface area of the reactant
  4. Temperature of the reaction
  5. Presence of a catalyst

EVALUATION

1. For a first order reaction $A \longrightarrow B$ the rate constant is x min^{-1}. If the initial concentration of A is 0.01M , the concentration of A after one hour is given by the expression.

   a) $0.01\ e^{-x}$ b) $1\times10^{-2}\left(1-e^{-60x}\right)$

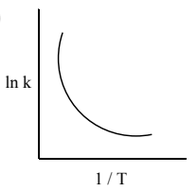
   c) $\left(1\times10^{-2}\right)e^{-60x}$ d) none of these

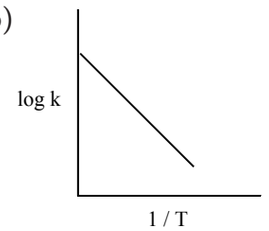
2. A zero order reaction $X \longrightarrow Product$, with an initial concentration 0.02M has a half life of 10 min. if one starts with concentration 0.04M, then the half life is

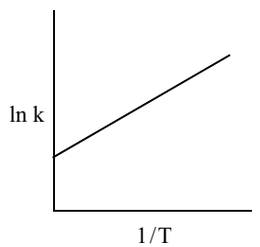
   a) 10 s b) 5 min c) 20 min

   d) cannot be predicted using the given information

3. Among the following graphs showing variation of rate constant with temperature (T) for a reaction, the one that exhibits Arrhenius behavior over the entire temperature range is

a)


b)


c)


d) both (b) and (c)

4. For a first order reaction $A \longrightarrow$ product with initial concentration x mol L^{-1} , has a half life period of 2.5 hours. For the same reaction with initial concentration $\left(\frac{x}{2}\right)$ mol L^{-1} the half life is

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